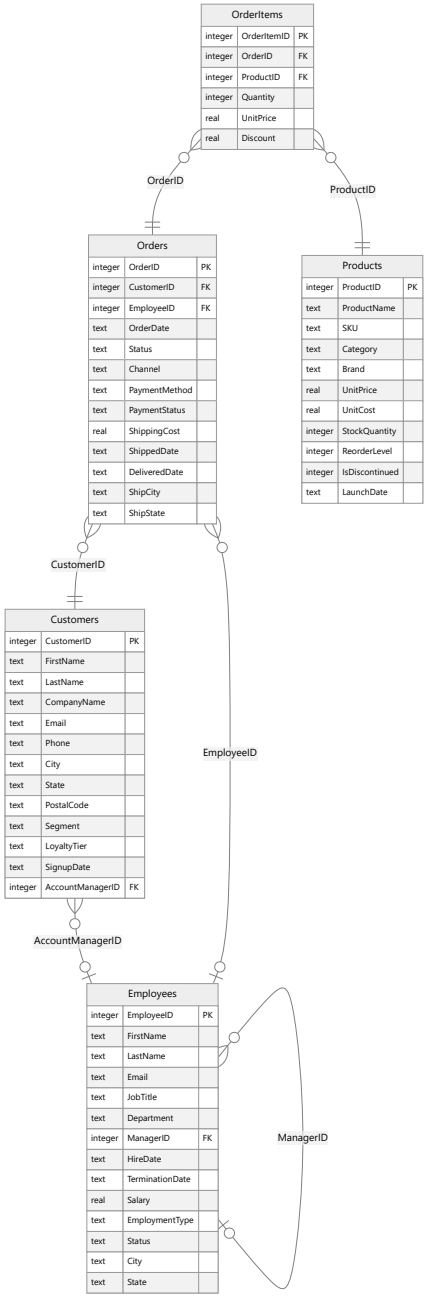
Sample database

Summit Supply Co. is a US omnichannel retailer selling electronics, computers, office supplies, furniture, home & kitchen, sports, books and health products to consumers and businesses. The database covers customers, orders and order lines, the product catalog, and the workforce.

- **Engine:** SQLite (`backend/data/company.db`) · **Snapshot date:** 2026-10-20 · **DDL:** `backend/data/schema.sql`
- **Generated by:** `backend/scripts/generate_data.py` (deterministic, fixed seed) · **Excel copy:** `backend/data/company_sample.xlsx`
- **Meaning of every column** comes from `backend/knowledge_base/semantic_layer.yaml` ; statistics and values are profiled from the live database.

Entity–relationship diagram



Tables

Table	Rows	Grain	Description
Employees	333	One row per employee (former employees are kept with Status 'Terminated').	Everyone who works or has worked at Summit Supply Co., including the org chart.
Customers	3,000	One row per customer account.	People and businesses who have an account with Summit Supply Co.
Products	111	One row per product (SKU).	The product catalog.
Orders	16,613	One row per order. Line items are in OrderItems.	Customer orders (one header row per order), including payment and fulfilment status.
OrderItems	30,449	One row per product per order.	Order lines — which products were bought in each order, how many, and at what price.

Employees

Everyone who works or has worked at Summit Supply Co., including the org chart.

Column	Type	Key	Description	Values / range
`EmployeeID`	INTEGER	PK	Unique employee identifier.	
`FirstName`	TEXT		Employee's first name.	
`LastName`	TEXT		Employee's last name.	
`Email`	TEXT		Work email address (unique).	
`JobTitle`	TEXT		Current job title.	53 distinct values
`Department`	TEXT		Department the employee belongs to.	`Sales`, `Engineering`, `Customer Support`, `Operations`, `Marketing`, `Finance`, `Product`, `Human Resources`, `Executive`
`ManagerID`	INTEGER	FK → Employees.EmployeeID	EmployeeID of the employee's direct manager; NULL only for the CEO.	
`HireDate`	TEXT		Date the employee joined.	2014-03-03 – 2026-10-17
`TerminationDate`	TEXT		Last day of employment; NULL for current employees.	2021-03-20 – 2026-10-17
`Salary`	REAL		Annual base salary in USD (pro-rated for part-time staff).	23000.0 – 340000.0
`EmploymentType`	TEXT		Contract type.	`Full-Time`, `Part-Time`, `Contract`
`Status`	TEXT		Employment status.	`Active`, `Terminated`, `On Leave`
`City`	TEXT		City the employee works from (office city, or home city for remote staff).	21 distinct values
`State`	TEXT		2-letter USPS code of the state the employee works from.	2-letter USPS state codes

Indexes: `idx_employees_manager` (ManagerID)

Customers

People and businesses who have an account with Summit Supply Co.

Column	Type	Key	Description	Values / range
`CustomerID`	INTEGER	PK	Unique customer identifier.	
`FirstName`	TEXT		First name of the customer (or the business's main contact).	
`LastName`	TEXT		Last name of the customer (or the business's main contact).	

Column	Type	Key	Description	Values / range
`CompanyName`	TEXT		Business name for Small Business and Corporate customers; NULL for consumers.	
`Email`	TEXT		Contact email address (unique).	
`Phone`	TEXT		Contact phone number; NULL when the customer did not provide one.	
`City`	TEXT		City of the customer's address.	
`State`	TEXT		2-letter USPS code of the state the customer lives in.	2-letter USPS state codes
`PostalCode`	TEXT		ZIP code of the customer's address.	
`Segment`	TEXT		Customer type.	`Consumer`, `Small Business`, `Corporate`
`LoyaltyTier`	TEXT		Loyalty tier based on lifetime spend on delivered orders.	`Bronze`, `Silver`, `Gold`, `Platinum`
`SignupDate`	TEXT		Date the customer created their account.	2021-06-01 – 2026-10-19
`AccountManagerID`	INTEGER	FK → Employees.EmployeeID	Employee assigned as the customer's account manager (Small Business and Corporate customers only; NULL for consumers).	

Products

The product catalog.

Column	Type	Key	Description	Values / range
`ProductID`	INTEGER	PK	Unique product identifier.	
`ProductName`	TEXT		Product name including its brand.	
`SKU`	TEXT		Stock-keeping unit code (unique), prefixed by a category code.	
`Category`	TEXT		Product category.	`Computers & Accessories`, `Electronics`, `Office Supplies`, `Furniture`, `Health & Personal Care`, `Home & Kitchen`, `Sports & Outdoors`, `Books`
`Brand`	TEXT		Brand name.	18 distinct values
`UnitPrice`	REAL		Current list price in USD (what it would sell for today, before discounts).	5.99 – 1512.99
`UnitCost`	REAL		Current unit cost to Summit Supply in USD (used for profit and inventory value).	3.2 – 673.27
`StockQuantity`	INTEGER		Units currently in stock.	0 – 1116
`ReorderLevel`	INTEGER		Stock level at or below which the product should be reordered.	10 – 100
`IsDiscontinued`	INTEGER		1 if the product is no longer sold, 0 if it is still sold.	`0`, `1`
`LaunchDate`	TEXT		Date the product was first available for sale.	2019-01-09 – 2026-10-16

Orders

Customer orders (one header row per order), including payment and fulfilment status.

Column	Type	Key	Description	Values / range
`OrderID`	INTEGER	PK	Unique order identifier.	
`CustomerID`	INTEGER	FK → Customers.CustomerID	The customer who placed the order.	

Column	Type	Key	Description	Values / range
`EmployeeID`	INTEGER	FK → Employees.EmployeeID	The employee (sales rep) who booked the order; NULL for self-service online orders from consumers.	
`OrderDate`	TEXT		Date the order was placed.	2022-01-03 – 2026-10-20
`Status`	TEXT		Fulfilment status of the order as of the snapshot date.	`Delivered`, `Cancelled`, `Returned`, `Shipped`, `Pending`, `Processing`
`Channel`	TEXT		Sales channel the order came through.	`Online`, `Phone`, `Partner`, `In-Store`
`PaymentMethod`	TEXT		How the order was paid for.	`Credit Card`, `Bank Transfer`, `PayPal`, `Digital Wallet`, `Gift Card`
`PaymentStatus`	TEXT		Payment state of the order as of the snapshot date.	`Paid`, `Refunded`, `Voided`, `Pending`
`ShippingCost`	REAL		Shipping fee charged to the customer in USD (0 for free shipping or in-store).	0.0 – 19.99
`ShippedDate`	TEXT		Date the order left the warehouse; NULL if not shipped yet or cancelled.	2022-01-04 – 2026-10-20
`DeliveredDate`	TEXT		Date the order was delivered; NULL if not delivered (yet).	2022-01-07 – 2026-10-20
`ShipCity`	TEXT		Delivery destination city.	
`ShipState`	TEXT		2-letter USPS code of the delivery destination state.	2-letter USPS state codes

Indexes: `idx_orders_employee` (EmployeeID), `idx_orders_orderdate` (OrderDate), `idx_orders_customer` (CustomerID)

OrderItems

Order lines — which products were bought in each order, how many, and at what price.

Column	Type	Key	Description	Values / range
`OrderItemID`	INTEGER	PK	Unique order-line identifier.	
`OrderID`	INTEGER	FK → Orders.OrderID	The order this line belongs to.	
`ProductID`	INTEGER	FK → Products.ProductID	The product bought on this line.	
`Quantity`	INTEGER		Units bought.	1 – 59
`UnitPrice`	REAL		Price per unit charged at the time of sale, before the line discount (USD).	5.15 – 1554.15
`Discount`	REAL		Discount on this line as a fraction (0.10 = 10% off).	0.0 – 0.3

Indexes: `idx_orderitems_product` (ProductID), `idx_orderitems_order` (OrderID)

Relationships

Join	Meaning
`Employees.ManagerID = Employees.EmployeeID`	Org chart — the employee's direct manager (self-join on Employees).
`Customers.AccountManagerID = Employees.EmployeeID`	The employee who manages this business customer's account.
`Orders.EmployeeID = Employees.EmployeeID`	The sales rep who booked the order.
`Orders.CustomerID = Customers.CustomerID`	The customer who placed the order.
`OrderItems.ProductID = Products.ProductID`	The product on an order line.
`OrderItems.OrderID = Orders.OrderID`	The order a line belongs to.

Customers ↔ Employees can be linked two ways — the customer's **account manager** (`Customers.AccountManagerID`) or the **sales rep who booked an order** (`Orders.EmployeeID`). The knowledge graph flags this ambiguity and the planner states which link it chose.

Business metrics

Validated definitions the agent reuses (each one was executed against the database when the knowledge base was built).

Metric	Definition	Value at snapshot
Net Revenue — Line value after discounts, excluding cancelled and returned orders and shipping.	`SUM(oi.Quantity * oi.UnitPrice * (1 - oi.Discount)) WHERE o.Status NOT IN ('Cancelled', 'Returned')`	20,259,520.22
Gross Revenue — Line value before discounts, excluding cancelled and returned orders.	`SUM(oi.Quantity * oi.UnitPrice) WHERE o.Status NOT IN ('Cancelled', 'Returned')`	21,488,663.04
Discount Amount — Money given away through line discounts on non-cancelled, non-returned orders.	`SUM(oi.Quantity * oi.UnitPrice * oi.Discount) WHERE o.Status NOT IN ('Cancelled', 'Returned')`	1,229,142.82
Units Sold — Units on non-cancelled, non-returned orders.	`SUM(oi.Quantity) WHERE o.Status NOT IN ('Cancelled', 'Returned')`	116,088
Order Count — Number of orders of any status. Exclude 'Cancelled' when the user says successful / valid orders.	`COUNT(DISTINCT o.OrderID)`	16,613
Average Order Value — Net revenue divided by the number of non-cancelled, non-returned orders.	`SUM(oi.Quantity * oi.UnitPrice * (1 - oi.Discount)) / COUNT(DISTINCT o.OrderID) WHERE o.Status NOT IN ('Cancelled', 'Returned')`	1,309.6
Gross Profit — Net revenue minus product cost (current UnitCost) on non-cancelled, non-returned orders.	`SUM(oi.Quantity * (oi.UnitPrice * (1 - oi.Discount) - p.UnitCost)) WHERE o.Status NOT IN ('Cancelled', 'Returned')`	7,958,207.64
Gross Margin % — Gross profit as a percentage of net revenue.	`100.0 * SUM(oi.Quantity * (oi.UnitPrice * (1 - oi.Discount) - p.UnitCost)) / SUM(oi.Quantity * oi.UnitPrice * (1 - oi.Discount)) WHERE o.Status NOT IN ('Cancelled', 'Returned')`	39.28
Cancellation Rate % — Share of all orders that were cancelled.	`100.0 * SUM(CASE WHEN o.Status = 'Cancelled' THEN 1 ELSE 0 END) / COUNT(*)`	3.89
Return Rate % — Share of delivered orders that were returned.	`100.0 * SUM(CASE WHEN o.Status = 'Returned' THEN 1 ELSE 0 END) / SUM(CASE WHEN o.Status IN ('Delivered', 'Returned') THEN 1 ELSE 0 END)`	3.13
Average Days to Ship — Average days between order and shipment, over shipped orders.	`AVG(julianday(o.ShippedDate) - julianday(o.OrderDate)) WHERE o.ShippedDate IS NOT NULL`	2.32
Average Days to Deliver — Average days between order and delivery, over delivered orders.	`AVG(julianday(o.DeliveredDate) - julianday(o.OrderDate)) WHERE o.DeliveredDate IS NOT NULL`	5.86
Customer Lifetime Value — Net revenue per customer who has bought at least once.	`SUM(oi.Quantity * oi.UnitPrice * (1 - oi.Discount)) / COUNT(DISTINCT o.CustomerID) WHERE o.Status NOT IN ('Cancelled', 'Returned')`	9,301.89
Outstanding Payments — Value billed (lines after discount plus shipping) on orders whose payment is still Pending.	`SELECT SUM(t.billed) AS value FROM ( SELECT o.OrderID, SUM(oi.Quantity * oi.UnitPrice * (1 - oi.Discount)) + o.ShippingCost AS billed FROM Orders o JOIN OrderItems oi ON oi.OrderID = o.OrderID WHERE o.PaymentStatus = 'Pending' GROUP BY o.OrderID, o.ShippingCost) t`	524,728.76
Repeat Customer Rate % — Share of buying customers with two or more non-cancelled orders.	`SELECT 100.0 * SUM(CASE WHEN n >= 2 THEN 1 ELSE 0 END) / COUNT(*) AS value FROM ( SELECT o.CustomerID, COUNT(*) AS n FROM Orders o WHERE o.Status <> 'Cancelled' GROUP BY o.CustomerID) t`	81.08
Headcount — Current employees (Active or On Leave).	`COUNT(*) WHERE e.Status IN ('Active', 'On Leave')`	287
Average Salary — Average annual salary of current employees.	`AVG(e.Salary) WHERE e.Status IN ('Active', 'On Leave')`	106,266.55
Total Payroll — Sum of annual salaries of current employees.	`SUM(e.Salary) WHERE e.Status IN ('Active', 'On Leave')`	30,498,500.0

Metric	Definition	Value at snapshot
Attrition Rate % — Share of everyone ever employed who has left. Filter TerminationDate for a specific period.	`100.0 * SUM(CASE WHEN e.Status = 'Terminated' THEN 1 ELSE 0 END) / COUNT(*)`	13.81
Average Tenure (years) — Average years since hire for current employees, as of the snapshot date.	`AVG((julianday('2026-10-20') - julianday(e.HireDate)) / 365.25) WHERE e.Status IN ('Active', 'On Leave')`	4.15
Inventory Value — Units in stock times unit cost, for products still sold.	`SUM(p.StockQuantity * p.UnitCost) WHERE p.IsDiscontinued = 0`	2,543,269.48

Business glossary

Term	Definition	SQL condition
active customer	A customer with at least one non-cancelled order in the 12 months before the snapshot date.	`EXISTS (SELECT 1 FROM Orders o WHERE o.CustomerID = c.CustomerID AND o.Status <> 'Cancelled' AND o.OrderDate > date('2026-10-20', '-12 months'))`
churned customer	A customer who has ordered before but not in the 12 months before the snapshot date.	`EXISTS (SELECT 1 FROM Orders o WHERE o.CustomerID = c.CustomerID) AND NOT EXISTS (SELECT 1 FROM Orders o WHERE o.CustomerID = c.CustomerID AND o.OrderDate > date('2026-10-20', '-12 months'))`
new customer	A customer whose SignupDate falls in the period being asked about.	`c.SignupDate >= :period_start`
repeat customer	A customer with two or more non-cancelled orders.	`(SELECT COUNT(*) FROM Orders o WHERE o.CustomerID = c.CustomerID AND o.Status <> 'Cancelled') >= 2`
customers without orders	Customers with no rows in Orders.	`NOT EXISTS (SELECT 1 FROM Orders o WHERE o.CustomerID = c.CustomerID)`
b2b	Small Business and Corporate customers.	`c.Segment IN ('Small Business', 'Corporate')`
b2c	Consumer customers.	`c.Segment = 'Consumer'`
top customers	Customers ranked by a measure — by Net Revenue unless the user names another (order count, units).	default: rank by Net Revenue, descending
best-selling products	Products ranked by Units Sold unless the user asks for revenue.	default: rank by Units Sold, descending
high-value order	An order whose net line value is at least $1,000.	`(SELECT SUM(oi.Quantity * oi.UnitPrice * (1 - oi.Discount)) FROM OrderItems oi WHERE oi.OrderID = o.OrderID) >= 1000`
open orders	Orders not yet delivered, cancelled or returned.	`o.Status IN ('Pending', 'Processing', 'Shipped')`
late shipment	An order shipped more than 3 days after it was placed.	`julianday(o.ShippedDate) - julianday(o.OrderDate) > 3`
overdue invoice	A bank-transfer order still unpaid more than 45 days after it was placed.	`o.PaymentMethod = 'Bank Transfer' AND o.PaymentStatus = 'Pending' AND o.OrderDate < date('2026-10-20', '-45 days')`
weekend orders	Orders placed on Saturday or Sunday.	`strftime('%w', o.OrderDate) IN ('0', '6')`
current employees	Employees with Status 'Active' or 'On Leave'.	`e.Status IN ('Active', 'On Leave')`
new hire	An employee hired in the 90 days before the snapshot date (or in the period asked about).	`e.HireDate > date('2026-10-20', '-90 days')`
manager	An employee with at least one direct report (someone whose ManagerID points to them). Note that JobTitle containing 'Manager' is a different, narrower test.	`EXISTS (SELECT 1 FROM Employees r WHERE r.ManagerID = e.EmployeeID)`
low stock	A product still sold whose stock is at or below its reorder level.	`p.IsDiscontinued = 0 AND p.StockQuantity <= p.ReorderLevel`
out of stock	A product with zero units in stock.	`p.StockQuantity = 0`

Term	Definition	SQL condition
products never ordered	Products that appear on no order line.	`NOT EXISTS (SELECT 1 FROM OrderItems oi WHERE oi.ProductID = p.ProductID)`

Conventions the agent follows

- Dates are ISO-8601 TEXT 'YYYY-MM-DD' with no time part. Compare them with string literals (OrderDate >= '2024-01-01'), bucket with strftime('%Y', col) / strftime('%Y-%m', col), and compute day differences with julianday(a) - julianday(b).
- The data is a snapshot as of 2026-10-20. Resolve relative periods ("last month", "this year", "last 90 days") against '2026-10-20' — e.g. date('2026-10-20', '-90 days') — not date('now').
- The fiscal year is the calendar year; quarters are calendar quarters (Q1 = Jan–Mar).
- State columns (Customers.State, Employees.State, Orders.ShipState) hold 2-letter USPS codes such as 'CA' or 'NY', never full names. "California" must become 'CA'.
- "Revenue" / "sales" means net revenue: SUM(OrderItems.Quantity * OrderItems.UnitPrice * (1 - OrderItems.Discount)) over orders whose Status is not 'Cancelled' or 'Returned', excluding shipping — unless the user explicitly asks to include those.
- OrderItems.UnitPrice is the price charged at the time of sale; Products.UnitPrice is today's list price. Always use OrderItems.UnitPrice for sales figures.
- Orders has no total column. An order's value is the sum of its lines (Quantity * UnitPrice * (1 - Discount)); add Orders.ShippingCost only when asked for the amount billed.
- OrderItems.Discount is a fraction from 0 to 0.30 (0.10 = 10% off), not a percentage.
- "Customers from X" filters Customers.State (where the customer lives). "Orders shipped to X" filters Orders.ShipState (delivery destination). About 10% of orders ship to a different state.
- Customers link to Employees in two different ways: Customers.AccountManagerID is the customer's assigned account manager (business customers only); Orders.EmployeeID is the sales rep who booked a particular order (NULL for self-service online consumer orders). Pick the one the question means.
- Current employees have Status IN ('Active', 'On Leave'). Former employees keep their row with Status 'Terminated' and a TerminationDate.
- Some customers have never ordered and some products have never sold. Use LEFT JOIN when the question asks to include them, or NOT EXISTS / LEFT JOIN ... IS NULL to find them.
- Products.IsDiscontinued is an INTEGER flag (1 = discontinued, 0 = still sold).